

Machine Learning Project

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Summary

- Introduction & Business case
- Dataset analysis
- Obstacles and solutions
- Results and comparison

Introduction

- **Diabetes:**

- **1921:** Major medical breakthrough: discovery of insulin
- No complete cure today
- ➔ Early detection = essential

Business case

- **Project objective:** Build a reliable and precise ML model using patients' features
 - ➔ Prevent severe complications
 - ➔ Reduce healthcare costs
- **Dataset:** 100k clinical records from Kaggle
- **Relevant to Data & AI major:** classification, preprocessing, optimization.

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Dataset Analysis – Quality

- MIT Certification
- Usability = 10
- Completeness = 100% (subtitles, tags, description & cover image)
- Credibility score = Compatible = 100%



Dataset Analysis – Features

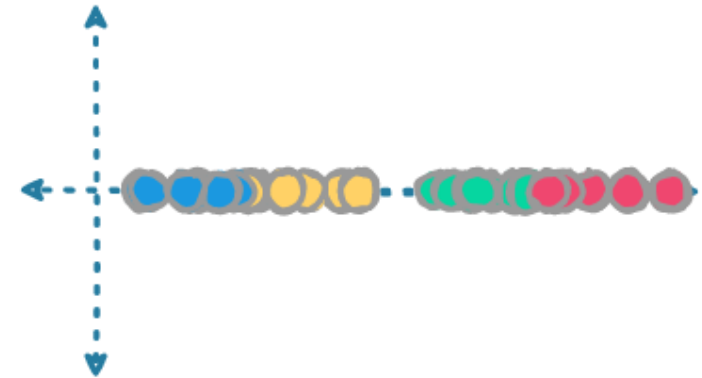
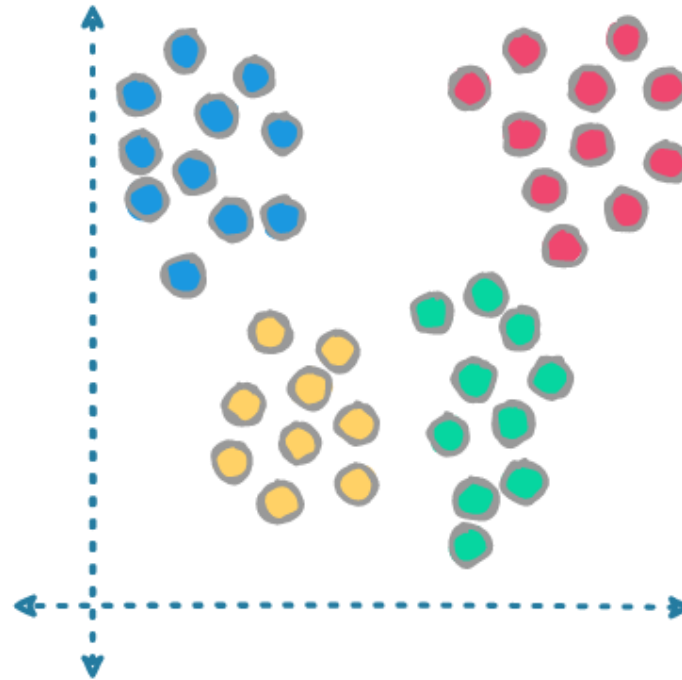
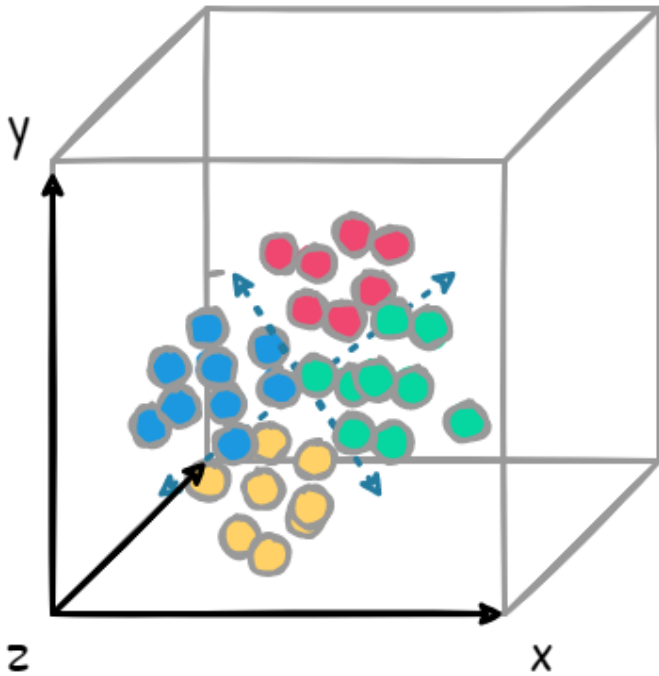
Feature	Role
year	Year of the diabetes test on the patient
gender	Gender of the patient (male, female or other)
age	Age of the patient
location	Geographical location of the patient (state, district...)
race: {AfricanAmerican, Asian, Caucasian...}	Origin of the patient (binary variables)
hypertension, heart_disease, diabetes	1 = Disease, 0 = no disease
smoking_history	Smoking frequency $\in$ {Never, Not Current, former...}
bmi	$\frac{weight}{height^2} \in [6.7 \text{ kg/m}^2, 98 \text{ kg/m}^2]$
hbA1c_level	Hemoglobin % of sugar in the blood
blood_glucose_level	Level of blood glucose

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Obstacles and solutions

Dimensionality Reduction



To SMOTE, or not to SMOTE?

- **Problem** : Oversampling:
 - Improves minority probability predictions
 - Worsens majority and overall log loss.
 - Only helps very weak classifiers like MLP.
- **Solution**: Good metrics, use a weighted loss (e.g. `scale_pos_weight` or `class_weight` in `lightgbm`), parameter tuning.

To SMOTE, or not to SMOTE?

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ABSTRACT

Balancing the data before training a classifier is a popular technique to address the challenges of imbalanced binary classification in tabular data. Balancing is commonly achieved by duplication of minority samples or by generation of synthetic minority samples. While it is well known that balancing affects each classifier differently, most prior empirical studies did not include strong state-of-the-art (SOTA) classifiers as baselines. In this work, we are interested in understanding whether balancing is beneficial, particularly in the context of SOTA classifiers. Thus, we conduct extensive experiments considering three SOTA classifiers along the weaker learners used in previous investigations. Additionally, we carefully discern proper metrics, consistent and non-consistent algorithms and hyper-parameter selection methods and show that these have a significant impact on prediction quality and on the effectiveness of balancing. Our results support the known utility of balancing for weak classifiers. However, we find that balancing does not improve prediction performance for the strong ones. We further identify several other scenarios for which balancing is effective and observe that prior studies demonstrated the utility of balancing by focusing on these settings.

CCS CONCEPTS

• Computing methodologies → Supervised learning by classification; Boosting.

KEYWORDS

binary classification, imbalance, SMOTE, balancing, oversampling

1 INTRODUCTION

In binary classification problems the classifier is tasked with classifying each sample to one of two classes. When the number of samples in the majority (bigger) class is considerably larger than the number of samples in the minority (smaller) class, the dataset is considered imbalanced. This skewness is believed to be challenging as both the classifier training process and the common metrics used to assess classification quality tend to be biased towards the majority class.

Numerous metrics have been proposed to address the challenges of imbalanced binary classification, e.g., Brier score, the area under the receiver operating characteristic curve (AUC), F_β and AM; see the survey of [42]. They are all non-symmetric and associate a higher loss with classifying a minority sample wrong compared to a majority sample. These metrics can be separated into two groups based on the underlying predictions used to compute them: 1) *Probability metrics* are calculated using the predicted class probabilities (e.g., AUC and Brier score); and 2) *label metrics* are based on the

predicted labels (e.g., f_β , balanced-accuracy and Jaccard similarity coefficient).

Machine learning classifiers typically optimize a symmetric objective function which associates the same loss with minority and majority samples. Thus, considering imbalanced classification problems, as previously noted in [4, 18], there seems to be a discrepancy between the classifier's symmetric optimization and the non-symmetric metric of interest which might result in an undesirable bias in the final trained model.

On the other hand, theoretical investigations into proper metrics and consistent algorithms suggest that the aforementioned discrepancy might not hinder prediction performance after all; see Section 2. In summary, these theoretical results state that when considering proper metrics or consistent classifiers, optimizing a symmetric objective function (commonly *logloss*) results in optimizing the non-symmetric metric of interest, suggesting that there is no need to address the data imbalance. Furthermore, some studies propose that skewed class distribution is not the only (or even the main) factor that hinders prediction quality for imbalanced datasets but the difficulty arises from other phenomena that are common in imbalanced data. These other contributing factors could be small sample size, class separability issues, noisy data, data drift and more [25, 37, 38].

In light of these theoretical results and ideas, we revisit the empirical evaluation of the techniques proposed to address the challenges of imbalanced data. These techniques could be roughly divided to three: 1) *preprocessing (or balancing)*, that augment the data, usually to be more balanced, before training the classifier; 2) *specialized classifiers*, that are specific for imbalanced data [11, 24, 34]. In many cases, this is achieved by increasing the weight of the minority samples in the objective function [8]; and 3) *postprocessing*, where the classifier predictions are augmented during inference [19, 31]. Due to its vast popularity, we focus on reevaluating the utility of balancing schemes.

The simplest balancing methods are either oversampling the minority class by duplicating minority samples or under-sampling the majority class by randomly removing majority samples. The idea of adding synthetic minority samples to tabular data was first proposed in SMOTE [6], where synthetic minority samples are created by interpolating pairs of the original minority points. SMOTE is extremely popular: 156 Google Scholar papers and 60 dblp papers with the word "SMOTE" in the title were published in 2021, 87 Stackoverflow questions discussing SMOTE were asked and dozens of blog posts demonstrating how to use it were published in 2021. Additionally, more than 100 SMOTE extensions and variants were proposed. See [10] for a comprehensive survey of SMOTE-like methods and [40] for additional statistics demonstrating the popularity of balancing in the community.

arXiv:2201.08528v3 [cs.LG] 11 May 2022

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Results for Baseline Models

Ranking	Model	Precision	Recall	F1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
	Log. Reg.	0.86	0.63	0.73	0.96	0.96	0.81	256	943	9686
	Dec. Tree	0.97	0.68	0.8	0.97	0.97	0.87	59	815	8209
	Rand. For.	1	0.67	0.8	0.97	0.96	0.85	8	846	8468
2	Gradient	0.98	0.68	0.81	0.97	0.98	0.88	38	805	8088
1	XGB	0.99	0.68	0.8	0.97	0.98	0.88	20	819	8210

Results for RandomOverSampling Models

Ranking	Model	Precision	Recall	F1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
	Log. Reg.	0.41	0.87	0.56	0.88	0.96	0.81	3157	327	6427
	Dec. Tree	0.44	0.9	0.59	0.89	0.97	0.86	2923	258	5503
	Rand. For.	0.91	0.7	0.79	0.97	0.97	0.86	183	758	7763
1	Gradient	0.47	0.91	0.62	0.91	0.98	0.88	2627	217	4797
2	XGB	0.48	0.91	0.63	0.91	0.98	0.88	2458	241	4868

Results for RandomUnderSampling Models

Ranking	Model	Precision	Recall	F1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
	Log. Reg.	0.41	0.87	0.56	0.88	0.96	0.81	3207	321	6417
	Dec. Tree	0.42	0.90	0.57	0.89	0.97	0.86	3152	252	5672
	Rand. For.	0.45	0.90	0.60	0.90	0.97	0.85	2837	246	5297
1	Gradient	0.46	0.91	0.61	0.90	0.98	0.88	2703	220	4903
2	XGB	0.46	0.91	0.61	0.90	0.98	0.87	2740	219	4930

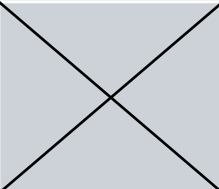
Results for SMOTE Models

Ranking	Model	Precision	Recall	F1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
	Log. Reg.	0.77	0.65	0.70	0.95	0.96	0.78	480	905	9530
	Dec. Tree	0.55	0.8	0.65	0.93	0.96	0.83	1671	520	6871
	Rand. For.	0.9	0.69	0.78	0.97	0.96	0.84	206	787	8076
1	Gradient	0.85	0.70	0.77	0.96	0.97	0.85	313	756	7873
2	XGB	0.92	0.69	0.79	0.97	0.97	0.86	160	787	8030

Results for StandardScaler Models

Ranking	Model	Precision	Recall	F1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
1	Log. Reg.	0.42	0.87	0.56	0.88	0.96	0.81	3135	326	6395
	SVC	0.43	0.85	0.57	0.89	0.95	0.77	2889	380	6689
	KNN	0.92	0.11	0.20	0.92	0.86	0.55	24	2258	22604

Results for PCA Models

Ranking	Model	Precision	Recall	F_1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
1	Log. Reg.	0.41	0.86	0.55	0.88	0.96	0.79	3208	355	6758
2	SVC	0.42	0.85	0.56	0.89	0.95	0.76	3008	384	6848
	KNN	0.92	0.17	0.28	0.93	0.87	0.59	39	2129	21329

Results for Grid Search Models

Ranking	Method	Precision	Recall	F1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
	Gradient	1	0.37	0.54	0.95	0.95	0.84	0	1605	16050
1	XGB	0.88	0.7	0.78	0.97	0.97	0.86	250	753	7780
	Log. Reg.	0.86	0.63	0.73	0.96	0.96	0.81	255	942	9675

Results for Voting, Bagging & Stacking

Ranking	Method	Precision	Recall	F1	Accuracy	AUC-ROC	AUC-PR	FP	FN	Custom Cost
	Voting	0.97	0.69	0.80	0.97	0.98	0.88	63	798	8043
	Bagging	0.92	0.69	0.79	0.97	0.97	0.86	150	784	7990
1	Stacking	0.91	0.71	0.80	0.97	0.98	0.88	171	739	7561

Global results: GridSearch on the Stacking

GradientBoostingRegressor
(learning_rate = 0.05,
min_samples_leaf = 10,
min_samples_split = 20,
n_estimators = 300,
random_state = 42)



THANK YOU

FOR

LISTENING